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Amaravati Quantum Valley Hackathon 2025

AMARAVATI QUANTUM VALLEY HACKATHON 2025

- **Problem Statement ID -** AQVH919
- **Problem Statement Title-** Logistics: Fleet Optimization
- **Theme-** Quantum-Driven Dynamic Fleet Routing
- **PS Category-** Software
- **Team ID-** U0037@202305103
- **Team Name -** Entangled Minds

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Quantum-Driven Vehicle Path Optimization in Logistics Networks

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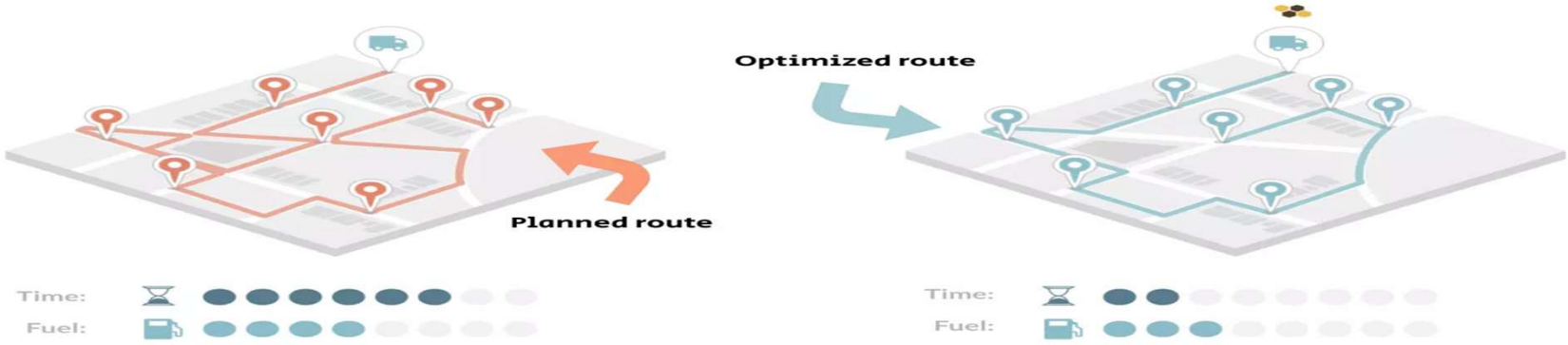
Proposed Solution

Transform traditional vehicle routing by adopting a hybrid quantum-classical framework for dynamic, real-time optimization. Moving beyond static plans, the system continuously adapts to traffic, orders, and vehicle status for optimal efficiency.

How the Quantum-Enhanced Routing Solution Addresses the Vehicle Routing Problem

- Combines quantum and classical techniques for faster route computation.
- Allows real-time updates by integrating live traffic, orders, and vehicle data.
- Utilizes quantum annealing to quickly optimize routes amid disruptions.
- Provides a scalable, predictive framework paving the way for autonomous logistic



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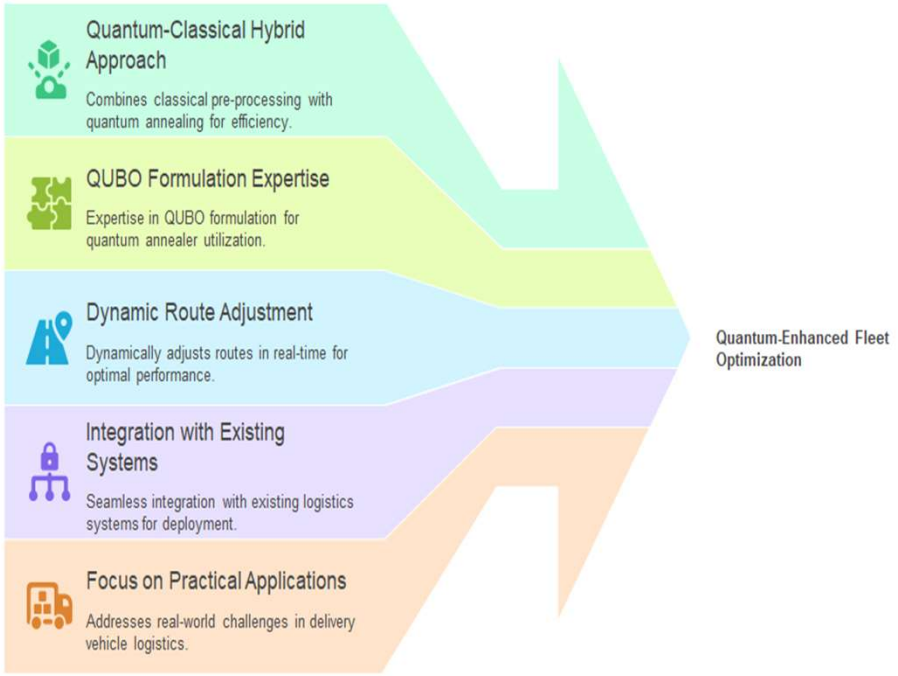
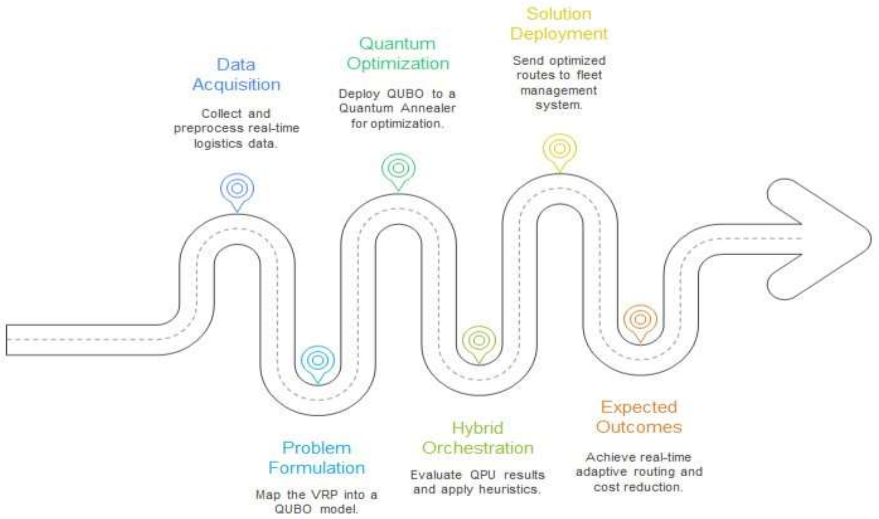
Quantum-Driven Logistics Innovation



TECHNICAL APPROACH



Languages: Python
Frameworks: Quantum computing SDK: D-Wave Ocean SDK,Classical ML, Tensor Flow, scikit Learn,Hybrid Work flows
Hardware/Cloud Platforms: Quantum Annealer, Classical Servers

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FEASIBILITY, VIABILITY & IMPACTS, BENEFITS

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Feasibility

Hybrid Systems

Practical applications of quantum computing are currently best achieved through hybrid quantum-classical systems, which combine the strengths of both approaches.

Pilot Projects

Early success has been demonstrated, such as Volkswagen's project using a quantum computer to optimize traffic flow in Beijing, showing the potential for handling large datasets and complex problems.

Algorithm Development

Advances in quantum optimization algorithms, which find the best solutions from many possibilities, are crucial for making this feasible.

Viability

Complex Optimization

Quantum computers are well-suited to solve the complex combinatorial optimization problems inherent in logistics, such as the Vehicle Routing Problem (VRP).

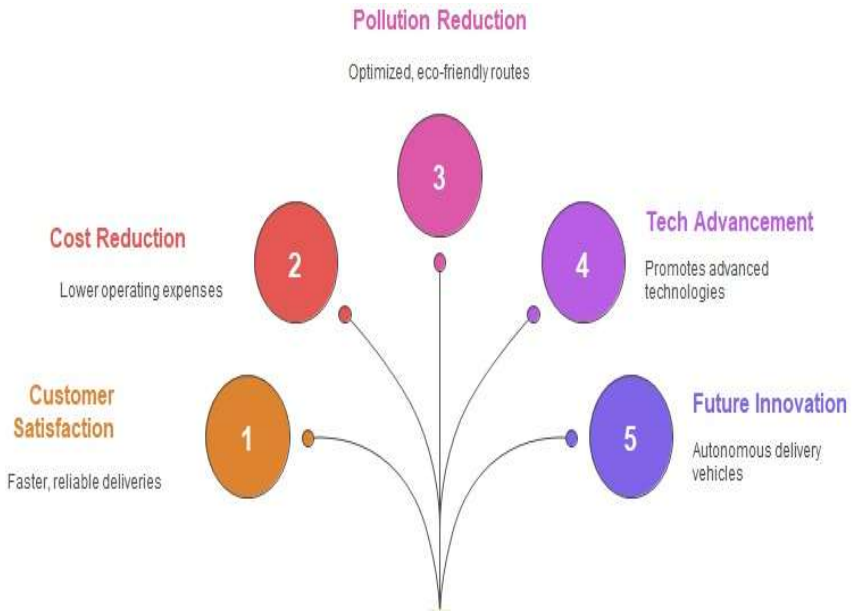
Scalability

As quantum hardware improves, the viability of solving even larger and more intricate logistics networks increases.

Long-Term Investment

Companies are beginning to develop strategies around quantum computing for operational and R&D advantages, signaling its growing long-term viability.

Impacts & Benefits



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SAMPLE WORKING OUTPUTS

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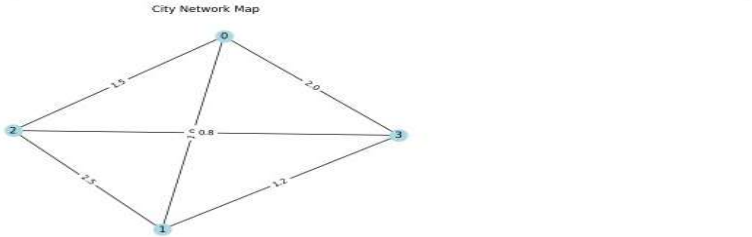
Based on Quantum annealer simulation & QUBO

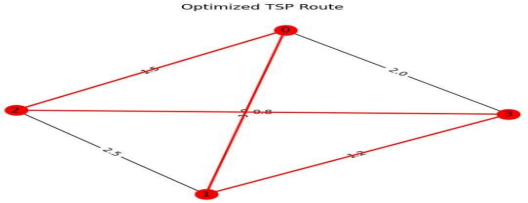
```
pip install dimod dwave-networkx
```

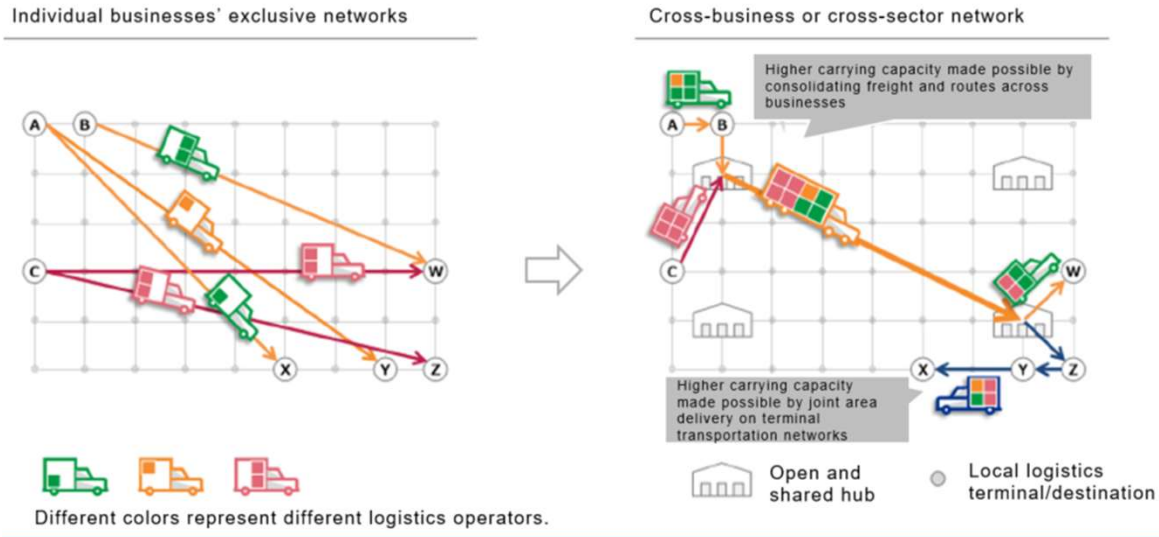
```
print("QUBO Dictionary (sample of first 10 terms):")
print(list(qubo.items())[:10])
print("\nNumber of QUBO variables (requires {} qubits):".format(len(qubo)))

# 3. Solve the QUBO using a Simulated Annealer
# dimod's SimulatedAnnealingSampler mimics the behavior of a quantum annealer
sampler = dimod.SimulatedAnnealingSampler()

# Run the sampler. num_reads is the number of times to run the annealing process.
sampleset = sampler.sample_qubo(qubo, num_reads=1000)

# 4. Interpret and Print the Results
# Get the best sample (lowest energy solution)
best_sample = sampleset.first.sample
energy = sampleset.first.energy

print("\nBest sample found:", best_sample)
print("Energy (cost) of solution:", energy)

# 5. Decode the solution back into a route
# The sample uses binary variables x_{city, position in tour}
# We need to find which city is at each position.
route = [None] * len(G) # Create an empty list for the route

# Iterate through all the variables in the best sample
for node, value in best_sample.items():
    # The variable name is a tuple: (city, time_step)
    if value == 1:
        city, time_step = node
        route[time_step] = city

# The route is a list of cities in the order they are visited
print("\nOptimized Route Order:", route)
# Since it's a cycle, we start and end at the same city. Let's complete the cycle.
route.append(route[0])
print("Complete Cycle Route:", route)

# 6. Calculate the total distance of the found route
total_distance = 0
for i in range(len(route)-1):
    city_1 = route[i]
    city_2 = route[i+1]
    total_distance += G[city_1][city_2]['weight']

print("Total Distance of the Route:", total_distance)

# 7. (Optional) Plot the optimized route
plt.figure()
nx.draw(G, pos, with_labels=True, node_color='lightblue')
nx.draw_networkx_edge_labels(G, pos, edge_labels=labels)

# Draw the path of the route
route_edges = [(route[i], route[i+1]) for i in range(len(route)-1)]
nx.draw_networkx_edges(G, pos, edgelist=route_edges, edge_color='red', width=2)
nx.draw_networkx_nodes(G, pos, nodelist=route, node_color='red')

plt.title("Optimized TSP Route")
plt.show()
```




```
QUBO Dictionary (sample of first 10 terms):
[((0, 0), (0, 0)), -12.0), (((0, 0), (0, 1)), 12.0), (((0, 0), (0, 2)), 12.0), (((0, 0), (0, 3)), 12.0), (((0, 1), (0, 1)), -12.0), (((0, 1), (0, 2)), 12.0), (((0, 1), (0, 3)), 12.0), (((0, 2), (0, 2)), -12.0), (((0, 2), (0, 3)), 12.0), (((0, 3), (0, 3)), -12.0)]

Number of QUBO variables (requires 136 qubits):
```

```
QUBO Dictionary (sample of first 10 terms):
[((0, 0), (0, 0)), -12.0), (((0, 0), (0, 1)), 12.0), (((0, 0), (0, 2)), 12.0), (((0, 0), (0, 3)), 12.0), (((0, 1), (0, 1)), -12.0), (((0, 1), (0, 2)), 12.0), (((0, 1), (0, 3)), 12.0), (((0, 2), (0, 2)), -12.0), (((0, 2), (0, 3)), 12.0), (((0, 3), (0, 3)), -12.0)]

Number of QUBO variables (requires 136 qubits):

Best sample found: {(0, 0): np.int8(0), (0, 1): np.int8(0), (0, 2): np.int8(1), (0, 3): np.int8(0), (1, 0): np.int8(0), (1, 1): np.int8(1), (1, 2): np.int8(0), (1, 3): np.int8(0), (2, 0): np.int8(0), (2, 1): np.int8(0), (2, 2): np.int8(0), (2, 3): np.int8(1), (3, 0): np.int8(1), (3, 1): np.int8(0), (3, 2): np.int8(0), (3, 3): np.int8(0)}
Energy (cost) of solution: -43.5

Optimized Route Order: [3, 1, 0, 2]
Complete Cycle Route: [3, 1, 0, 2, 3]
Total Distance of the Route: 4.5
```





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RESEARCH AND REFERENCES

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Key Research Papers/Articles:

- “Quantum algorithms for vehicle routing problems”
- “Quantum annealing for solving vehicle routing problems”
- “The Vehicle Routing Problem”
- “Quantum Annealing for Vehicle Routing and Fleet Management” https://ieeexplore.ieee.org/document/8756014
- “Hybrid Quantum-Classical Optimization for Fleet Routing”